7. Write a Servlet program to store a user's preferences (like theme or language) using cookies. Retrieve and display these preferences on subsequent visits.

AIM

To develop a Java Servlet-based web application that stores user preferences (such as theme and language) using cookies and retrieves them on subsequent visits to personalize the weather information webpage.

ALGORITHM

1. Create a **HTML login page** to accept username and password.
2. Develop a **LoginServlet** to validate user credentials.
3. If valid, create a session using:

```
HttpSession session = request.getSession();
```

4. Store user data in session:

```
session.setAttribute("user", username);
```

5. Redirect user to **DashboardServlet**.
6. In DashboardServlet, retrieve session using:

```
request.getSession(false);
```

7. Display session details (user, session ID, time).
8. Create a **LogoutServlet** to invalidate session:

```
session.invalidate();
```

9. Redirect back to login page after logout.

SOURCE CODE

1. HTML Page (index.html)

```
<!DOCTYPE html>
<html>
<head>
<title>Preferences</title>
<style>
body { font-family: Arial; text-align: center; margin-top: 50px; }
button { margin: 5px; padding: 8px; }
</style>
</head>

<body>

<h2>Set Preferences</h2>

<!-- Theme -->
```

```
<p>Theme:</p>
<button onclick="setPref('theme','dark')">Dark</button>
<button onclick="setPref('theme','light')">Light</button>

<!-- Language -->
<p>Language:</p>
<button onclick="setPref('lang','English')">English</button>
<button onclick="setPref('lang','Tamil')">Tamil</button>

<br><br>
<button onclick="save()">Save</button>
<button onclick="clearCookies()">Clear</button>

<h3 id="output"></h3>

<script>
let prefs = {};

function setPref(key, val) {
  prefs[key] = val;
}

function setCookie(k, v) {
  document.cookie = k + "=" + v + "; path=/";
}

function getCookies() {
  return document.cookie;
}

function save() {
  for (let k in prefs) setCookie(k, prefs[k]);
  output.innerText = "Saved: " + getCookies();
}

function clearCookies() {
  document.cookie = "theme=; expires=Thu, 01 Jan 1970 UTC;";
  document.cookie = "lang=; expires=Thu, 01 Jan 1970 UTC;";
  output.innerText = "Cookies Cleared";
}
</script>

</body>
</html>
```

2. PreferenceServlet.java

```
import java.io.; import javax.servlet.; import javax.servlet.http.*;

public class PreferenceServlet extends HttpServlet {

protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    String theme = request.getParameter("theme");
    String language = request.getParameter("language");

    // Create cookies
    Cookie themeCookie = new Cookie("theme", theme);
    Cookie languageCookie = new Cookie("language", language);

    // Set expiry time (1 day)
    themeCookie.setMaxAge(24 * 60 * 60);
    languageCookie.setMaxAge(24 * 60 * 60);
```

```
    // Add cookies to response
    response.addCookie(themeCookie);
    response.addCookie(languageCookie);

    PrintWriter out = response.getWriter();
    out.println("<html><body>");
    out.println("<h3>Preferences Saved Successfully!</h3>");
    out.println("<a href='DisplayServlet'>Go to Weather Page</a>");
    out.println("</body></html>");
}


}
```

3. DisplayServlet.java

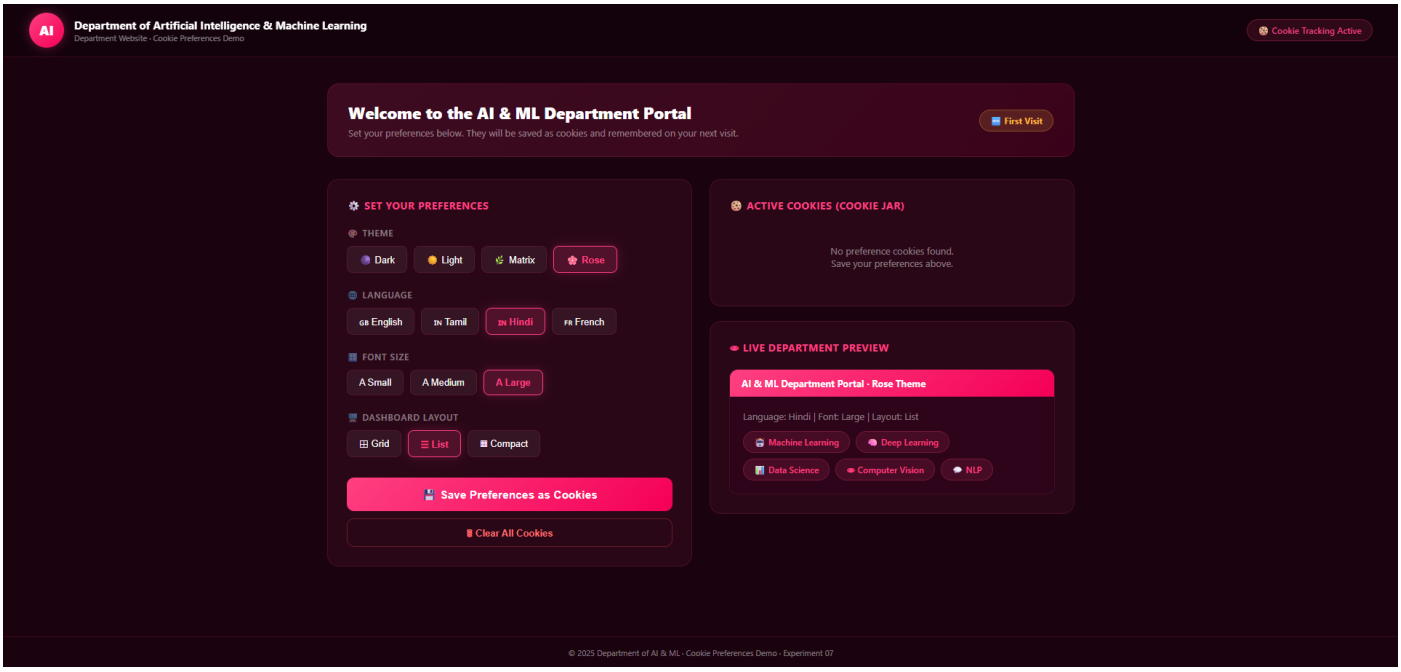
```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/logout")
public class LogoutServlet extends HttpServlet {
   protected void doGet(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {

      HttpSession session = request.getSession(false);
      if (session != null) {
         session.invalidate();
      }
      response.sendRedirect("login.html");
   }
}
```

4. web.xml

```
<web-app>

   <servlet>
      <servlet-name>PreferenceServlet</servlet-name>
      <servlet-class>PreferenceServlet</servlet-class>
   </servlet>

   <servlet-mapping>
      <servlet-name>PreferenceServlet</servlet-name>
      <url-pattern>/PreferenceServlet</url-pattern>
   </servlet-mapping>

   <servlet>
      <servlet-name>DisplayServlet</servlet-name>
      <servlet-class>DisplayServlet</servlet-class>
   </servlet>
```

```
    <servlet-mapping>
        <servlet-name>DisplayServlet</servlet-name>
        <url-pattern>/DisplayServlet</url-pattern>
    </servlet-mapping>

</web-app>
```

OUTPUT :



Result:

The experiment was successfully completed.User preferences (theme and language) were stored using cookies and retrieved on subsequent visits, allowing the AI-ML webpage to provide a personalized user experience.